A variational criterion for Gaussian-preserving Wasserstein flows

Let
$$R : \mathcal{P}_2(\mathbb{R}^d) \to \mathcal{P}_2(\mathbb{R}^d)$$
be the map sending a probability measure to the Gaussian measure with the same mean and covariance:
$$R\mu := \mathsf{N}(m_\mu, \Sigma_\mu),$$
where
$$m_\mu := \int_{\mathbb{R}^d} x \, \mathrm{d}\mu(x), \qquad \Sigma_\mu := \int_{\mathbb{R}^d} (x - m_\mu)(x - m_\mu)^\mathsf{T} \, \mathrm{d}\mu(x).$$
The global condition suggested by Hugo Lavenant is
$$F(R\mu) \leq F(\mu) \qquad \text{for every } \mu \in \mathcal{P}_2(\mathbb{R}^d).$$
This condition is a clean sufficient condition for the JKO scheme, and hence for the Wasserstein gradient flow when the latter is uniquely obtained as the JKO limit, to preserve Gaussian measures.

It is important, however, to distinguish this global condition from mere continuous-time Gaussian preservation. The condition
$$F(R\mu) \leq F(\mu)$$
is stronger than saying that the Wasserstein gradient flow preserves Gaussian initial data. A simple smooth cylinder functional gives a counterexample to the converse.

1 Gelbrich contraction

Let
$$\mathcal{G}_d := \{\mathsf{N}(m, \Sigma) : m \in \mathbb{R}^d, \ \Sigma \in \mathbb{S}^d_+\}$$
be the family of possibly degenerate Gaussian laws.

For $A, B \in \mathbb{S}^d_+$, define
$$\mathfrak{B}^2(A, B) := \operatorname{Tr} A + \operatorname{Tr} B - 2 \operatorname{Tr}\Big((A^{1/2} B A^{1/2})^{1/2}\Big).$$

Theorem 1 (Gelbrich's inequality)**.** *For every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*
$$W_2^2(\mu, \nu) \geq \|m_\mu - m_\nu\|^2 + \mathfrak{B}^2(\Sigma_\mu, \Sigma_\nu).$$
Moreover, for Gaussian laws,
$$W_2^2(\mathsf{N}(m, A), \mathsf{N}(n, B)) = \|m - n\|^2 + \mathfrak{B}^2(A, B).$$
Consequently,
$$W_2(R\mu, R\nu) \leq W_2(\mu, \nu) \qquad \text{for every } \mu, \nu \in \mathcal{P}_2(\mathbb{R}^d).$$

Proof. The first inequality is Gelbrich's inequality. Applying the Gaussian formula to $R\mu$ and $R\nu$ gives

$$W_2^2(R\mu, R\nu) = \|m_\mu - m_\nu\|^2 + \mathfrak{B}^2(\Sigma_\mu, \Sigma_\nu).$$

Gelbrich's inequality then yields

$$W_2^2(R\mu, R\nu) \le W_2^2(\mu, \nu).$$

□

2 Lavenant's variational criterion

Let $F : \mathcal{P}_2(\mathbb{R}^d) \to (-\infty, +\infty]$ be a functional. For $\tau > 0$ and $\gamma \in \mathcal{P}_2(\mathbb{R}^d)$, define the JKO objective

$$\mathcal{J}_{\tau,\gamma}(\eta) := F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta).$$

Theorem 2 (Lavenant's sufficient condition)**.** *Assume*

$$F(R\mu) \le F(\mu) \qquad \text{for every } \mu \in \mathcal{P}_2(\mathbb{R}^d).$$

Let $\gamma \in \mathcal{G}_d$. Then, for every $\eta \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{J}_{\tau,\gamma}(R\eta) \le \mathcal{J}_{\tau,\gamma}(\eta).$$

Consequently, if the JKO problem

$$\inf_{\eta \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta) \right\}$$

admits a minimizer, then it admits a Gaussian minimizer. If the minimizer is unique, then it is Gaussian.

Proof. Since γ is Gaussian,

$$R\gamma = \gamma.$$

By Gelbrich contraction,

$$W_2(R\eta, \gamma) = W_2(R\eta, R\gamma) \le W_2(\eta, \gamma).$$

By assumption,

$$F(R\eta) \le F(\eta).$$

Therefore

$$F(R\eta) + \frac{1}{2\tau} W_2^2(\gamma, R\eta) \le F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta).$$

Thus projecting any competitor through R can only decrease the JKO objective.

If η is a minimizer, then $R\eta$ is also a minimizer and is Gaussian. If the minimizer is unique, then $\eta = R\eta$, hence η is Gaussian. □

Corollary 1. *Assume*

$$F(R\mu) \le F(\mu) \qquad \text{for every } \mu \in \mathcal{P}_2(\mathbb{R}^d).$$

If the Wasserstein gradient flow of F is unique and obtained as the limit of the JKO scheme, then Gaussian initial data remain Gaussian.

3 The exact variational equivalence

The condition

$$F(R\mu) \le F(\mu)$$

is not merely sufficient for the previous proof. It is exactly equivalent to the fact that Gaussianization improves every JKO objective with Gaussian base point.

Theorem 3. *The following two statements are equivalent.*

1. *For every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$,*
$$F(R\mu) \le F(\mu).$$
2. *For every $\tau > 0$, every $\gamma \in \mathcal{G}_d$, and every $\eta \in \mathcal{P}_2(\mathbb{R}^d)$,*
$$\mathcal{J}_{\tau,\gamma}(R\eta) \le \mathcal{J}_{\tau,\gamma}(\eta).$$

Proof. Assume first that

$$F(R\mu) \le F(\mu)$$

for all μ. If γ is Gaussian, then $R\gamma = \gamma$, and by Gelbrich contraction,

$$W_2(\gamma, R\eta) = W_2(R\gamma, R\eta) \le W_2(\gamma, \eta).$$

Therefore

$$F(R\eta) + \frac{1}{2\tau} W_2^2(\gamma, R\eta) \le F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta).$$

Conversely, assume that the JKO comparison holds. Fix $\eta \in \mathcal{P}_2(\mathbb{R}^d)$ and choose

$$\gamma := R\eta.$$

Then γ is Gaussian, and the comparison gives

$$F(R\eta) \le F(\eta) + \frac{1}{2\tau} W_2^2(R\eta, \eta).$$

Letting $\tau \to +\infty$ yields

$$F(R\eta) \le F(\eta).$$

$\square$

Remark 1. *Thus the condition*

$$F \circ R \le F$$

is the right condition for the specific variational argument based on the retraction R. What is not automatic is the converse implication from continuous-time Gaussian preservation to this global inequality.

4 Examples satisfying $F \circ R \le F$

Example 1 (Functionals of mean and covariance)**.** *Suppose*

$$F(\mu) = \Phi(m_\mu, \Sigma_\mu)$$

for some map

$$\Phi : \mathbb{R}^d \times \mathbb{S}_+^d \to (-\infty, +\infty].$$

Then

$$F(R\mu) = F(\mu).$$

In particular, quadratic interaction energies are included. If

$$F(\mu) = \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} (x-y)^\mathsf{T} Q (x-y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y),$$

with $Q = Q^\mathsf{T}$, then

$$F(\mu) = \mathrm{Tr}(Q\Sigma_\mu),$$

so F depends only on the covariance.

Proof. The identities

$$m_{R\mu} = m_\mu, \qquad \Sigma_{R\mu} = \Sigma_\mu$$

give the first claim.

For the quadratic interaction, let X, Y be independent random variables with law μ. Then

$$\mathbb{E}[(X-Y)(X-Y)^\mathsf{T}] = 2\Sigma_\mu.$$

Hence

$$F(\mu) = \frac{1}{2} \mathrm{Tr}\Big(Q \, \mathbb{E}[(X-Y)(X-Y)^\mathsf{T}]\Big) = \mathrm{Tr}(Q\Sigma_\mu).$$

□

Example 2 (Boltzmann entropy)**.** *Define*

$$\mathrm{Ent}(\mu) := \begin{cases} \displaystyle\int_{\mathbb{R}^d} \rho \log \rho \, \mathrm{d}x, & \mu = \rho \, \mathrm{d}x, \\ +\infty, & \textit{otherwise.} \end{cases}$$

Then

$$\mathrm{Ent}(R\mu) \le \mathrm{Ent}(\mu).$$

Proof. Assume first that $\mu = \rho \, \mathrm{d}x$ has finite entropy and nondegenerate covariance. Let g be the density of $R\mu$. The nonnegativity of relative entropy gives

$$0 \le \int_{\mathbb{R}^d} \rho \log \frac{\rho}{g} \, \mathrm{d}x = \mathrm{Ent}(\mu) - \int_{\mathbb{R}^d} \rho \log g \, \mathrm{d}x.$$

Since $\log g$ is a constant plus a quadratic polynomial, and since μ and $R\mu$ have the same mean and covariance,

$$\int_{\mathbb{R}^d} \rho \log g \, \mathrm{d}x = \int_{\mathbb{R}^d} g \log g \, \mathrm{d}x = \mathrm{Ent}(R\mu).$$

Therefore

$$\mathrm{Ent}(R\mu) \le \mathrm{Ent}(\mu).$$

The degenerate case follows by approximation or by restricting to the affine span of the covariance. □

Example 3 (Squared transport cost to a Gaussian)**.** *Let $\gamma \in \mathcal{G}_d$ and define*

$$F(\mu) := \frac{1}{2} W_2^2(\mu, \gamma).$$

Then

$$F(R\mu) \le F(\mu).$$

Proof. Since γ is Gaussian,

$$R\gamma = \gamma.$$

By Gelbrich contraction,

$$W_2(R\mu, \gamma) = W_2(R\mu, R\gamma) \le W_2(\mu, \gamma).$$

Squaring gives the claim. □

5 Why the converse is false without extra assumptions

One may ask whether Gaussian preservation of the continuous-time Wasserstein gradient flow implies

$$F(R\mu) \le F(\mu) \qquad \text{for all } \mu.$$

In this generality, the answer is no. The reason is that flow preservation is a local condition along the Gaussian manifold, whereas

$$F(R\mu) \le F(\mu)$$

is a global comparison between a measure and its Gaussianization.

Proposition 1 (A counterexample to necessity)**.** *There exists a smooth bounded functional $F : \mathcal{P}_2(\mathbb{R}) \to (0, 1]$ such that:*

1. *every Gaussian measure is a stationary point of the formal Wasserstein gradient flow of F;*
2. *hence, under uniqueness of the flow, Gaussian initial data remain Gaussian;*
3. *nevertheless, there exists $\mu \in \mathcal{P}_2(\mathbb{R})$ such that*

$$F(R\mu) > F(\mu).$$

Thus Gaussian preservation of the flow does not imply

$$F \circ R \le F.$$

Proof. For $\mu \in \mathcal{P}_2(\mathbb{R})$, write

$$m_\mu := \int x \, \mathrm{d}\mu(x), \qquad \sigma_\mu^2 := \int (x - m_\mu)^2 \, \mathrm{d}\mu(x).$$

Define

$$D(\mu) := \int_{\mathbb{R}} \cos x \, \mathrm{d}\mu(x) - e^{-\sigma_\mu^2/2} \cos(m_\mu),$$

and set

$$F(\mu) := \exp(-D(\mu)^2).$$

If $\mu = \mathsf{N}(m, \sigma^2)$, then

$$\int_{\mathbb{R}} \cos x \, \mathrm{d}\mu(x) = e^{-\sigma^2/2} \cos m.$$

Therefore

$$D(\mu) = 0 \qquad \text{for every Gaussian } \mu,$$

and hence

$$F(\mu) = 1 \qquad \text{for every Gaussian } \mu.$$

We next compute the Wasserstein differential. Let $(\mu_t)_t$ be a smooth curve solving

$$\partial_t \mu_t + \partial_x(\mu_t v_t) = 0.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} m_{\mu_t} = \int v_t \, \mathrm{d}\mu_t,$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t} \sigma_{\mu_t}^2 = 2 \int (x - m_{\mu_t}) v_t(x) \, \mathrm{d}\mu_t(x).$$

Also,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \cos x \ \mathrm{d}\mu_t(x) = \int -\sin x \, v_t(x) \ \mathrm{d}\mu_t(x).$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}t}D(\mu_t) = \int G_{\mu_t}(x)v_t(x) \ \mathrm{d}\mu_t(x),$$

where

$$G_\mu(x) = -\sin x + e^{-\sigma_\mu^2/2}\left[\sin(m_\mu) + \cos(m_\mu)(x - m_\mu)\right].$$

Consequently,

$$\nabla_W D(\mu)(x) = G_\mu(x),$$

and

$$\nabla_W F(\mu)(x) = -2D(\mu)e^{-D(\mu)^2}G_\mu(x).$$

If γ is Gaussian, then $D(\gamma) = 0$, so

$$\nabla_W F(\gamma) = 0.$$

Therefore every Gaussian is a stationary solution of the formal Wasserstein gradient-flow equation

$$\partial_t \mu_t = \partial_x \left(\mu_t \nabla_W F(\mu_t)\right).$$

Under uniqueness, the flow starting from a Gaussian remains at that Gaussian.

It remains to check that the inequality

$$F(R\mu) \leq F(\mu)$$

fails. Take

$$\mu := \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1.$$

Then

$$m_\mu = 0, \qquad \sigma_\mu^2 = 1, \qquad R\mu = \mathsf{N}(0,1).$$

Hence

$$D(R\mu) = 0,$$

while

$$D(\mu) = \cos(1) - e^{-1/2} \neq 0.$$

Therefore

$$F(R\mu) = 1$$

and

$$F(\mu) = \exp\left(-(\cos(1) - e^{-1/2})^2\right) < 1.$$

Thus

$$F(R\mu) > F(\mu),$$

which is the opposite of the desired inequality. □

Remark 2. *This counterexample is intentionally simple. It shows that a functional may have zero Wasserstein gradient on the whole Gaussian manifold, so that Gaussian measures are stationary, while assigning larger values to Gaussian measures than to some non-Gaussian measures with the same mean and covariance.*

6 Conclusion

The condition

$$F(R\mu) \leq F(\mu)$$

has a precise variational meaning: it is equivalent to saying that the Gaussianization map R improves every JKO objective with Gaussian base point. Together with Gelbrich's contraction, this gives a direct proof that the JKO scheme can be selected inside the Gaussian manifold.

However, this condition is not necessary for the weaker property that the continuous-time Wasserstein gradient flow preserves Gaussian initial data. Preservation of the flow is local along $\mathcal{G}_d$, whereas

$$F(R\mu) \leq F(\mu)$$

is a global comparison between μ and its moment-matched Gaussian.

References

[1] M. Gelbrich, On a formula for the L^2-Wasserstein metric between measures on Euclidean and Hilbert spaces, *Mathematische Nachrichten* **147** (1990), 185–203.

[2] H. Lavenant, Gaussian-preserving Wasserstein flows, personal communication, October 2025.